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Abstract

The increasing complexity and scale of cyber threats necessitate advanced and adaptive defense mechanisms. Traditional reinforcement learning (RL)-based network defense systems often face challenges related to centralized data collection, privacy concerns, and scalability. To address these issues, this paper proposes a conceptual framework for Federated Reinforcement Learning (FRL) in network defense systems. The framework enables decentralized learning by allowing multiple entities to collaboratively train a global model while preserving data privacy. Key components of the framework include secure data-sharing mechanisms, privacy-preserving communication protocols, and trust management strategies to ensure reliable collaboration among participating entities. By integrating federated learning (FL) with RL, the proposed approach enhances threat detection, anomaly response, and real-time adaptability in dynamic network environments. Additionally, this framework minimizes data exposure risks while maintaining model accuracy, making it suitable for large-scale, privacy-sensitive network defense applications. The paper further discusses the challenges and potential solutions associated with implementing FRL in cybersecurity, paving the way for a more robust and privacy-aware defense strategy.

Keywords (separated by '-')

Federated Reinforcement Learning - Network Defense - Privacy-Preserving Machine Learning - Adaptive Threat Detection

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Conceptual Framework for Federated Reinforcement Learning in Network Defense

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Abstract. The increasing complexity and scale of cyber threats necessitate advanced and adaptive defense mechanisms. Traditional reinforcement learning (RL)-based network defense systems often face challenges related to centralized data collection, privacy concerns, and scalability. AQ1 To address these issues, this paper proposes a conceptual framework for Federated Reinforcement Learning (FRL) in network defense systems. The framework enables decentralized learning by allowing multiple entities to collaboratively train a global model while preserving data privacy. AQ2 Key components of the framework include secure data-sharing mechanisms, privacy-preserving communication protocols, and trust management strategies to ensure reliable collaboration among participating entities. By integrating federated learning (FL) with RL, the proposed approach enhances threat detection, anomaly response, and real-time adaptability in dynamic network environments. Additionally, this framework minimizes data exposure risks while maintaining model accuracy, making it suitable for large-scale, privacy-sensitive network defense applications. The paper further discusses the challenges and potential solutions associated with implementing FRL in cybersecurity, paving the way for a more robust and privacy-aware defense strategy.

Keywords: Federated Reinforcement Learning · Network Defense · Privacy-Preserving Machine Learning · Adaptive Threat Detection

1 Introduction

In today's hyper-connected digital landscape, cybersecurity threats are evolving in both scale and sophistication. Traditional network defense mechanisms often struggle to keep pace with emerging attack vectors that exploit vulnerabilities across decentralized infrastructures. Reinforcement Learning (RL) has gained attention in recent years for its ability to learn optimal defense strategies through continuous interaction with dynamic environments. However, its dependency on centralized data collection and training poses critical challenges—especially in

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terms of data privacy, communication overhead, and scalability. To address these limitations, Federated Reinforcement Learning (FRL) has emerged as a promising paradigm that enables multiple entities to collaboratively train RL models without sharing raw data. By decentralizing the learning process, FRL significantly reduces privacy risks while maintaining model effectiveness. This approach is particularly relevant for modern networked systems such as IoT environments, multi-domain cloud architectures, and edge computing platforms—where data is distributed and sensitive [1]. AQ3

In this paper, we propose a conceptual framework for applying FRL in the context of network defense based on two experimental results. The framework emphasizes privacy-preserving learning, secure inter-agent communication, and robust trust management [2]. By structuring collaborative learning processes around federated principles, our framework enhances the system's ability to detect, adapt to, and mitigate cyber threats in real time—without compromising the confidentiality of the participating nodes' data. We also explore key components of the proposed framework, including data-sharing mechanisms [3], privacy-preserving protocols, and collaboration strategies. Furthermore, the paper discusses implementation challenges and outlines potential research directions for deploying FRL in real-world cybersecurity systems.

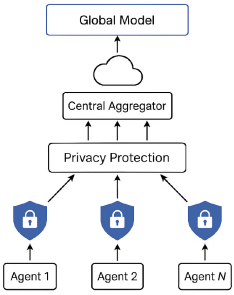

Fig. 1. General FRL Framework for Network Defense

The figure above, Fig. 1 is shown the General Architecture Framework including the general components of a Federated Reinforcement Learning Framework.

2 Hybrid Approach Combining Related Works

2.1 Hybrid Approach

Traditional RL frameworks rely heavily on centralized training, which introduces serious privacy concerns, especially when dealing with sensitive or regulated data. To mitigate these challenges, Federated Learning (FL) has emerged as a solution that enables decentralized model training without requiring raw data to

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be shared. FL has been widely adopted in domains such as mobile device personalization, healthcare, and finance. The application of FL in security, however, is still in its nascent stages. Studies such as [4] have demonstrated its potential in anomaly detection across distributed networks, but few address its synergy with reinforcement learning. Recent works have begun to explore Federated Reinforcement Learning (FRL) as a hybrid approach. For example, [5] introduced a distributed malware mitigation system using FRL, where agents learn defensive strategies based on local observations. Similarly, [6] proposed mechanisms for model protection in complementary environments. However, these studies often focus on algorithmic performance, with limited discussion on practical deployment concerns such as communication overhead, trust management, and privacy trade-offs.

Existing works generally optimize detection accuracy or communication efficiency, but they treat privacy as a static mechanism and trust as an external factor (e.g., client selection). Our proposed framework differs by **coupling trust and privacy**: a node's trust score continuously modulates its local differential privacy (DP) budget before uploading model updates. This trust-conditioned privacy schedule (DAPI) ensures that reliable nodes inject less noise, improving accuracy, while low-trust nodes are restricted with stronger privacy protection.

3 Privacy Preserving Techniques

Privacy preservation is central to the proposed Federated Reinforcement Learning (FRL) framework [7]. To ensure collaborative learning without compromising data confidentiality, we integrate a hybrid set of privacy-preserving methods, including Differential Privacy (DP), Homomorphic Encryption (HE), and Secure Aggregation protocols.

3.1 Hybrid Privacy Approach: DP + Lightweight Encryption

A hybrid strategy combines local noise injection via DP and secure transmission via lightweight encryption (e.g., Paillier cryptosystem) to balance accuracy, privacy, and communication overhead.

Differential Privacy (DP) is applied during local model training to prevent any single data point from significantly affecting the global model. The privacy budget ϵ controls this trade-off. A calibration mechanism is defined as:

$$\epsilon = \frac{\Delta}{\alpha} \cdot \lambda \tag{1}$$

where Δ is the gradient sensitivity, α is the target minimum accuracy, and λ is a tunable parameter [8]

Lightweight Encryption such as the Paillier cryptosystem allows encrypted gradients to be aggregated without decryption, preserving confidentiality in transmission [9].

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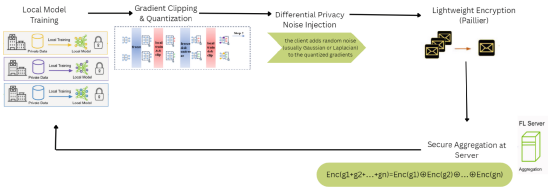

Fig. 2. schematic diagram illustrating the hybrid privacy pipeline

3.2 Pipeline Overview

Figure 2 illustrates the complete hybrid privacy pipeline [10] for Federated Reinforcement Learning (FRL)—starting from model updates on the client side to their secure aggregation on the server.

3.3 Comparative Evaluation Experiment 01

To evaluate privacy–accuracy trade-offs under realistic conditions, we used the **CIC-IDS2017** and **UNSW-NB15** network intrusion datasets, both widely used for federated intrusion detection research.Each dataset was partitioned across clients in a non-IID fashion(e.g., by day or by attack type). This setup reflects enterprise branches monitoring heterogeneous network environments. In Table 1 shown below;the comparison table includes the Secure Multi-Party Computation (SMPC) since Secure Aggregation outputs plaintext aggregates, it is combined with DP to mitigate inference risks.

Table 1. Privacy–Accuracy Trade-offs on CIC-IDS2017

Configuration	Acc. (%)	F1	ϵ	Comm. (MB)
DP ($\epsilon = 1.0$)	94.2	0.938	1.0	7.9
DP ($\epsilon = 0.5$)+Paillier	93.5	0.930	0.5	10.6
SecAgg only	95.1	0.946	–	8.1
DP+SecAgg ($\epsilon = 1.0$)	94.0	0.936	1.0	8.1
HE only (Paillier)	93.9	0.934	–	10.6
SMPC	94.7	0.942	–	9.8

DP ($\epsilon = 1.0$) – Moderate accuracy, strong privacy
DP ($\epsilon = 0.5$)+Paillier – Higher privacy, higher cost
SecAgg only – High performance, no DP guarantees
DP+SecAgg ($\epsilon = 1.0$) – Balanced trade-off
HE only (Paillier) – Strong security, heavy cost
SMPC – Collaborative secure aggregation

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4 Conceptual Framework

This section presents a high-level architecture of the proposed Federated Reinforcement Learning (FRL) framework for network defense while preserving privacy and minimizing communication costs.

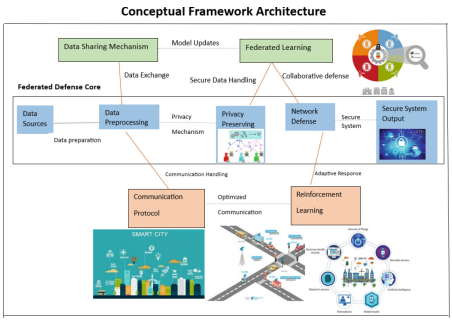

Fig. 3. Conceptual Framework Architecture

The framework is composed of five key components, as shown in Fig. 3 above.

4.1 Overview of Federated Reinforcement Learning in Network Defense

Federated Reinforcement Learning (FRL) merges the principles of Reinforcement Learning (RL) and Federated Learning (FL) to enable decentralized agents [11] to collaboratively learn optimal defense strategies without compromising the privacy of their local data. In the context of network defense, where sensitive data is distributed across multiple nodes, such as routers, firewalls, IoT devices or cloud endpoints, FRL offers a promising solution to build intelligent defense mechanisms that are adaptive and privacy aware [12,13].

Traditional RL relies on centralized training using a shared dataset or environment, which poses significant risks in privacy-sensitive settings. In contrast, FRL allows each agent to train its model locally based on its own observations and rewards from the environment. Periodically, these agents share model updates (rather than raw data) with a central aggregator, which combines the updates (e.g., using FedAvg or other aggregation rules) and redistributes a refined global model back to the agents.

This setup provides the best of both worlds:

Adaptability through reinforcement learning, where agents learn from interaction with cyber threats.

Privacy and scalability through federated learning, where data remains local, and only model parameters are shared.

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In a typical FRL-based network defense system [14], agents might represent different sub-networks or domains within a large organization. They collaborate to identify anomalies, block malicious IPs, or adjust firewall rules—learning continually while preserving data locality.

4.2 Data Sharing Mechanism

In a Federated Reinforcement Learning (FRL) environment, data sharing is replaced by model sharing—a key aspect that addresses privacy and data sovereignty [15] concerns in modern network defense systems. Each participating agent (e.g., a firewall, router, or IoT node) collects local observations from its environment and trains a reinforcement learning model using its own data [16]. Instead of sending raw logs or traffic patterns to a central location, agents periodically transmit their locally updated model parameters to a federated aggregator.

Federated reinforcement learning (FRL) enables decentralized model training, where each agent independently learns from locally observed states, actions, and rewards. Instead of sharing raw data, agents transmit only model updates, like weights or gradients, to a central server or peer coordinator. These updates are aggregated, often using methods like Federated Averaging (FedAvg), to refine the global model, which is then redistributed to all agents for further training.

This approach offers significant advantages, including enhanced data privacy by avoiding raw data transmission, reduced bandwidth requirements through compressed model updates, and personalized learning that adapts to each agent's unique environment, making it particularly effective for diverse threat landscapes.

4.3 Privacy Preserving

Privacy preservation is a cornerstone of the proposed Federated Reinforcement Learning (FRL) framework, particularly in the context of cybersecurity, where agents operate on highly sensitive data such as network logs, access patterns, and user behavior [17]. To ensure that the collaborative learning process does not compromise the confidentiality of individual entities, the framework integrates multiple privacy-preserving mechanisms.

Differential Privacy (DP) introduces calibrated statistical noise into model gradients or parameters before they are transmitted to the central aggregator. This ensures that the presence or absence of any individual data point does not significantly influence the final model. The main advantage of DP lies in its strong mathematical privacy guarantees. However, excessive noise can degrade model performance and learning accuracy.

Secure Multi-Party Computation (SMPC) allows agents to collaboratively compute global models without revealing their private inputs. In FRL, this is critical during model aggregation, where encrypted or partitioned parameters are combined securely, preventing both the central server and peer agents from accessing individual updates. The trade-off is higher computational and coordination overhead. In Section III-C we therefore extended the experimental

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comparison to include SMPC [18], making the trade-offs among DP, HE, SecAgg, and SMPC more intuitive.

Homomorphic Encryption (HE) enables computations directly on encrypted data[18], allowing agents to share encrypted model updates without exposing sensitive information. This provides strong protection but can be computationally and bandwidth intensive, which may limit real-time applications.

Together, these techniques form a privacy-preserving layer that supports collaborative learning while aligning with data protection regulations (e.g., GDPR). In practice, a hybrid integration can be used—for example, combining DP with lightweight secure aggregation—to maintain a balance between computational efficiency and privacy assurance. As shown in Table 1 in section III; subsection C. Each mechanism makes different trade-offs between accuracy, communication cost, and security strength. This comparative view clarifies why our framework adopts a flexible hybrid strategy.

4.4 Communication Protocols

The efficacy of a Federated Reinforcement Learning (FRL) framework for network defense depends not only on local learning and privacy preservation but also on the robustness[18] and efficiency of its communication protocols. In a distributed environment where agents exchange model updates with a central aggregator or among peers, secure and efficient communication is critical to ensure real-time collaboration and prevent adversarial interference.

To protect the integrity and confidentiality of model updates, all communications are conducted over encrypted channels, typically using Transport Layer Security (TLS). The use of mutual TLS (mTLS) further authenticates both communicating parties, ensuring that only trusted agents participate in the federation.

The FRL framework supports both synchronous and asynchronous update mechanisms. In the synchronous case, agents train locally and send updates in coordinated rounds, while in the asynchronous case, updates are sent independently, improving responsiveness but introducing the challenge of model staleness.

Communication reliability mechanisms can be expressed as follows. First, a retry mechanism is triggered if a client update is not acknowledged:

$$P_{retry}(i) = \begin{cases} 1, & \text{if } \mathrm{Ack}(i) = 0 \\ 0, & \text{if } \mathrm{Ack}(i) = 1 \end{cases} \tag{2}$$

where $\mathrm{Ack}(i)$ is the acknowledgement flag for client i.

Second, version control ensures only consistent updates are aggregated:

$$M^{(t+1)} = M^{(t)} + \frac{1}{N} \sum_{i=1}^{N} \Delta w_i^{(t)} \cdot \mathbb{1}\{v_i = v_{server}\}, \tag{3}$$

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where $\Delta w_i^{(t)}$ is the update from client i at round t, and $\mathbb{1}\{v_i = v_{server}\}$ filters out stale or mismatched versions.

Third, anomaly-based filtering is used to detect malicious or disconnected clients:

$$\mathcal{F} = \{\Delta w_i \mid \|\Delta w_i - \mu\|_2 \leq \tau\}, \tag{4}$$

where μ is the mean of received updates and τ is an anomaly threshold. Only updates within τ are aggregated, while outliers are discarded.

In addition, communication efficiency is improved through compression (quantization and sparsification), gradient clipping, and thresholding to transmit only significant parameter changes. These mechanisms collectively enable scalable, secure, and fault-tolerant collaboration across FRL agents, forming the backbone of the system's ability to adapt to evolving cyber threats while preserving operational stability.

Collectively, these communication protocols enable scalable, secure, and efficient collaboration across agents participating in the FRL framework. They form the backbone of the system's ability to adapt to evolving cyber threats while preserving operational stability and performance.

5 Federated Defense for Distributed Enterprise Networks

To demonstrate the applicability of the proposed Federated Reinforcement Learning (FRL) framework, this section is based on the second experiment, involving a large enterprise with multiple geographically distributed branches—each operating its own network infrastructure and monitoring systems. These branches face a wide range of cyber threats, including Distributed Denial-of-Service (DDoS) attacks, zero-day exploits, and insider threats. However, due to regulatory, legal, or internal policy constraints, direct sharing of raw data (e.g., system logs, user activity records) between branches is not feasible.

Using the proposed FRL framework, each branch deploys a local agent that monitors its network traffic, detects anomalies, and learns defensive policies through reinforcement learning. These local agents do not share sensitive logs or payload data. Instead, they periodically send encrypted or noise-injected model updates to a central aggregator that constructs a global defense policy. This policy is then redistributed to all agents, enabling them to benefit from collective learning while maintaining data confidentiality.

Over time, the agents learn to detect emerging attack patterns collaboratively. For instance, if a new malware variant is detected in one branch, the local model quickly adapts, and the knowledge is federated to other branches—even those that have not yet encountered the threat. Trust and collaboration are enforced through the framework's trust management system. If an agent consistently sends anomalous updates, its trust score is reduced, and it may be excluded from future rounds. This protects the system against poisoned models and ensures high-quality contributions. Additionally, the framework supports adaptive update frequencies based on local threat levels. A branch under active

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attack may trigger immediate sharing cycles to spread critical insights faster. This use case highlights the key strengths of the proposed framework—namely, decentralized intelligence, privacy preservation, rapid knowledge transfer, and adversarial resilience—making it highly suitable for modern distributed security environments.

5.1 Trust Enforcement Through Dynamic Trust Scores in FRL Systems

Trust management is a critical aspect of Federated Reinforcement Learning, ensuring that participating nodes act in the collective interest without compromising the security of the overall system. Effective trust enforcement dynamically adjusts the influence of each node based on its observed behavior and historical contributions, thereby mitigating the risk of malicious activities like model poisoning and data leakage. This approach enables more reliable global model aggregation and supports the long-term stability of the FRL system.

5.2 Use Case Scenario: Dynamic Adaptive Mechanisms

Trust and collaboration are enforced through the framework's trust management system. If an agent consistently sends anomalous updates, its trust score is reduced, and it may be excluded from future rounds. To further enhance this approach, a dynamic adaptive mechanism can be integrated that adjusts the intensity of privacy protection based on the historical contributions and trust scores of the nodes.

1) Dynamic Adaptive Privacy Intensity (DAPI) Mechanism: Dynamic Adaptive Privacy Intensity (DAPI) is a novel approach that adjusts privacy protection levels based on the trust score of each node. This mechanism aims to optimize the trade-off between privacy and model accuracy, providing lower noise levels to high-trust nodes while enhancing privacy for low-trust nodes. This ensures that more reliable contributors have a stronger influence on the global model, while potentially malicious nodes face stricter privacy measures, reducing the risk of model poisoning and backdoor attacks. The key components of DAPI Mechanism are :

Trust Score Calculation - Experiment 02
Trust scores are dynamically adjusted based on node behavior, historical contributions, and the quality of submitted updates. Factors influencing trust scores include:

- **Model Accuracy:** Consistency of local model performance.
- **Update Stability:** Variance in gradient updates.
- **Reputation Feedback:** Peer evaluations or anomaly detection.

Privacy Intensity Modulation
Nodes with higher trust scores receive lower levels of noise injection, while lower-trust nodes have stronger privacy protections. Noise levels are adjusted using a

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parameter ϵ (epsilon) in differential privacy (DP), where lower ϵ means stronger privacy.

The relationship can be modeled as:

$$\epsilon_i = \beta \cdot T_i + \gamma \tag{5}$$

where:

- β is a scaling factor that defines the sensitivity of the privacy level to trust changes.
- γ is a baseline privacy guarantee, ensuring even high-trust nodes retain a minimum level of privacy.

Example Parameter Settings:

AQ4 (See Table 2).

Table 2. Dynamic Privacy Intensity Based on Trust Scores

Trust Score (T_i)	Privacy Budget (ϵ_i)	Noise Intensity
0.2 (Low Trust)	0.5	High
0.5 (Moderate)	1.0	Medium
0.8 (High Trust)	2.0	Low

Integration Into FRL Framework

- **Local Model Training:** Nodes train locally with noise injection scaled according to their trust scores.
- **Model Update Transmission:** Nodes with higher trust transmit less noisy updates, enhancing global model accuracy.
- **Central Aggregation:** Aggregator uses the trust-modulated updates for more precise model merging.

Potential Advantages and Challenges

Advantages:

- Improved Model Accuracy: High-trust nodes contribute less noisy updates, enhancing global model quality.
- Resilience to Model Poisoning: Low-trust nodes face stronger privacy controls, reducing the impact of malicious contributions.
- Efficient Use of Bandwidth: Reduced noise for trusted nodes decreases data size and communication costs.

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Challenges:

- Continuous Trust Monitoring: Requires real-time evaluation of node behavior to adjust privacy budgets.
- Scalability: High computational overhead for large-scale networks.
- Adversarial Resilience: Potential for coordinated attacks to manipulate trust scores.

2) Visualization and Performance Analysis: To validate this approach, illustrating the relationship between trust scores and privacy budgets will be included in Fig. 4, along with comparative analysis tables to highlight performance improvements.

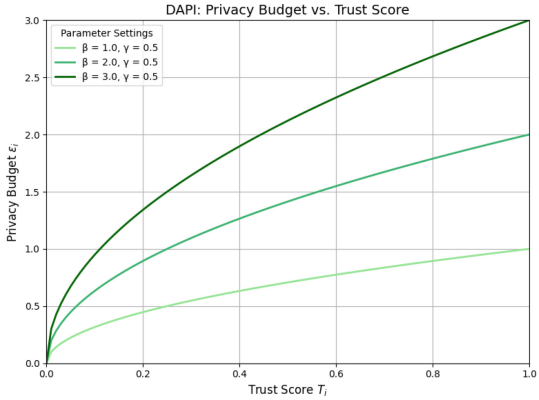

Fig. 4. Dynamic Adaptive Privacy Intensity (DAPI).

6 Challenges and Open Issues

While the proposed Federated Reinforcement Learning (FRL) framework offers significant potential for network defense, its deployment in real-world environments presents several open challenges that must be addressed to ensure scalability, robustness, and trustworthiness.

1) **Heterogeneity of Data and Environments:** Agents in a federated system may operate in vastly different environments with varying types of network traffic, hardware configurations, and threat landscapes. This heterogeneity can result in non-identically distributed (non-IID) data across nodes, which may slow convergence and reduce the effectiveness of the aggregated global model.
2) **Communication Overhead and Latency:** FRL systems still require frequent communication between agents and the aggregator. In bandwidth-limited or latency-sensitive environments, this can hinder real-time decision-making, especially during attack scenarios that demand rapid updates.

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3) **Model Poisoning and Adversarial Attacks:** Malicious clients may attempt to compromise the system by sending poisoned or manipulated updates. Trust management can partially address this issue, but more advanced Byzantine-resilient aggregation and anomaly detection algorithms are still required to guarantee robustness against sophisticated adversaries.
4) **Privacy vs. Accuracy Trade-off:** Techniques such as differential privacy, homomorphic encryption, and secure aggregation protect sensitive data, but often at the cost of reduced accuracy or increased computation. Balancing these trade-offs is essential, particularly in latency-sensitive environments where both compliance and operational performance are required.
5) **Incentive Design and Participation Dropouts:** Without formal incentive mechanisms or penalties for non-cooperation, nodes may drop out or contribute low-quality updates, reducing system stability.
6) **Interpretability and Auditing:** Developing explainable FRL frameworks that provide transparency and support auditing remains an important research direction.

7 Conclusion and Future Work

This paper has presented a conceptual framework for applying Federated Reinforcement Learning (FRL) in network defense, addressing critical challenges in cybersecurity such as data privacy, adaptability, and distributed threat response. By enabling decentralized agents to collaboratively learn optimal defense strategies without sharing sensitive raw data, the proposed framework ensures both model robustness and regulatory compliance. The integration of secure communication protocols, privacy-preserving techniques, and dynamic trust management enhances the system's ability to withstand adversarial conditions and scale across diverse environments.

Through its modular design and based on the experiment data, the framework accommodates the unique needs of real-world applications, including those in enterprise networks, critical infrastructure, and Internet-of-Things (IoT) ecosystems. It also provides a foundation for cross-domain collaboration where conventional centralized learning approaches fall short.

While promising, the framework also opens up a number of avenues for future research. These include improving defense against model poisoning attacks, optimizing communication efficiency in high-latency networks, and developing explainable federated reinforcement learning models for transparent decision-making. Furthermore, the incorporation of blockchain-based audit trails, incentive-driven agent participation, and hybrid hierarchical federations could extend the utility and trustworthiness of FRL in increasingly complex network defense scenarios.

Ultimately, this work contributes a comprehensive foundation for designing intelligent, secure, and privacy-aware collaborative learning systems capable of evolving with modern cyber threats.

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Chapter 17

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